Variation and selection

Original exam-style questions · 30 questions

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Question 1

core · 1 mark · 0610_T18_P1_20261018_Q1

Which comparison is an example of variation within a species?

A the difference between a fish and a flowering plant

B different body lengths among adult individuals of one fish species

C the difference between an animal cell and a bacterial cell

D the difference between water and soil

Question 2

core · 1 mark · 0610_T18_P1_20261018_Q2

Which measurement is most likely to show continuous variation in a population?

A ABO blood group

B whether pea seeds are smooth or wrinkled

C whether a plant has one of two inherited flower colours

D adult body mass

Question 3

core · 1 mark · 0610_T18_P1_20261018_Q3

Which observation is evidence of discontinuous variation?

A individuals fall into distinct groups with no intermediate phenotypes

B individuals have every possible length within a range

C all masses are arranged along a continuous scale

D values can be measured to more decimal places

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Question 4

core · 1 mark · 0610_T18_P1_20261018_Q4

Identical plant genotypes grown with different nutrient supplies reach different heights. Which conclusion is supported?

A All variation must be caused by mutation.

B Height can never be inherited.

C The environment can contribute to continuous variation.

D Each plant must have a different blood group.

Question 5

core · 1 mark · 0610_T18_P1_20261018_Q5

What is the original source of a new allele?

A training a muscle

B learning a behaviour

C normal feeding alone

D mutation

Question 6

core · 1 mark · 0610_T18_P1_20261018_Q6

Which exposure can increase mutation rate?

A a lower sound volume

B ionising radiation

C a different photoperiod in every case

D a reduction in visible colour contrast

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Question 7

supplement · 1 mark · 0610_T18_P1_20261018_Q7

Which event is a gene mutation?

A movement of an unchanged enzyme through cytoplasm

B the normal use of glucose in respiration

C a random change in a DNA base sequence

D a temporary increase in leaf temperature alone

Question 8

supplement · 1 mark · 0610_T18_P1_20261018_Q8

Which set contains only sources of genetic variation?

A mutation, meiosis and random fertilisation

B exercise, memory and training

C diet, injury and sunlight warming the skin

D respiration, digestion and sweating

Question 9

core · 1 mark · 0610_T18_P1_20261018_Q9

Which feature best fits the definition of an adaptive feature?

A any injury acquired during life

B an inherited feature increasing survival and reproduction in an environment

C any feature that makes an organism larger

D a change that always prevents reproduction

Question 10

core · 1 mark · 0610_T18_P1_20261018_Q10

A nocturnal desert mammal avoids daytime heat. What is the most reasonable benefit of this behaviour?

A elimination of all need for food

B conversion of the mammal into a producer

C reduced exposure to high temperatures and water loss

D prevention of all genetic variation

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Question 11

supplement · 1 mark · 0610_T18_P1_20261018_Q11

Which feature is most useful for reducing transpiration in a xerophyte?

A a very thin cuticle on every surface

B permanently wide-open stomata

C absence of any water storage in all conditions

D a thick waxy cuticle

Question 12

core · 1 mark · 0610_T18_P1_20261018_Q12

During natural selection, which individuals are most likely to contribute alleles to the next generation?

A those that survive and reproduce more successfully in that environment

B those that need a new feature most urgently

C those that always have the largest body

D those selected only by a farmer

Question 13

core · 1 mark · 0610_T18_P1_20261018_Q13

Which action is selective breeding?

A allowing every animal to mate without selection

B giving all animals antibiotics to create useful mutations

C crossing animals chosen for high milk yield and selecting high-yielding offspring

D changing an animal's genes by inserting a bacterial gene

Question 14

supplement · 1 mark · 0610_T18_P1_20261018_Q14

A bacterial population becomes resistant after repeated antibiotic treatment. Which explanation is best?

A resistant variants survive treatment and reproduce

B each bacterium chooses to become resistant

C the antibiotic is a nutrient that all bacteria need

D all susceptible bacteria survive better than resistant ones

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Question 15

supplement · 1 mark · 0610_T18_P1_20261018_Q15

What distinguishes artificial selection from natural selection?

A Artificial selection produces no inherited changes.

B Humans choose which individuals reproduce in artificial selection.

C Natural selection always requires human choice.

D Only artificial selection involves variation.

Question 16

core · 4 marks · 0610_T18_P1_20261018_Q16

A class measures the hand spans of adult volunteers from one population.

(a) Explain why these measurements are an example of variation and identify the type. [2]

(b) State two influences that can contribute to differences in this type of feature. [2]

Question 17

core · 4 marks · 0610_T18_P1_20261018_Q17

A survey assigns people to blood groups A, B, AB and O.

(a) Explain why these results show discontinuous variation. [2]

(b) Describe the usual cause of discontinuous variation and contrast it with continuous variation. [2]

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Question 18

core · 4 marks · 0610_T18_P1_20261018_Q18

A student measures 40 leaves. Lengths range from 18 mm to 61 mm.

(a) Calculate the range and identify the type of variation. [2]

(b) State two steps that improve the reliability of comparisons between plants. [2]

Question 19

core · 4 marks · 0610_T18_P1_20261018_Q19

A population acquires a new allele after a genetic change.

(a) Describe mutation and its role in forming alleles. [2]

(b) State two types of factor that can increase mutation rate. [2]

Question 20

supplement · 4 marks · 0610_T18_P1_20261018_Q20

Two siblings inherit different combinations of their parents' alleles.

(a) Describe a gene mutation. [2]

(b) State two processes, other than mutation, that contribute to genetic variation between siblings. [2]

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Question 21

core · 4 marks · 0610_T18_P1_20261018_Q21

Two groups of genetically identical seedlings are grown under different light intensities and develop different masses.

(a) Explain why their difference in mass does not prove a difference in genes. [2]

(b) Describe two controls needed when testing the effect of light alone. [2]

Question 22

core · 4 marks · 0610_T18_P1_20261018_Q22

Researchers record body length and seed shape in a pea population. Seed shape is either smooth or wrinkled.

(a) Describe how the two sets of phenotypes differ. [2]

(b) Explain how each set should be recorded accurately. [2]

Question 23

supplement · 4 marks · 0610_T18_P1_20261018_Q23

A species reproduces sexually. A new DNA sequence appears in one lineage.

(a) Explain which process can create the new sequence and describe that process. [2]

(b) Explain how sexual reproduction can increase variation without creating a new allele. [2]

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Question 24

supplement · 4 marks · 0610_T18_P1_20261018_Q24

A desert plant has rolled leaves and stomata in sunken pits.

(a) Explain how these two features reduce water loss. [2]

(b) Explain the benefit of a thick cuticle and water-storage tissue. [2]

Question 25

core · 4 marks · 0610_T18_P1_20261018_Q25

A water bird has broad webbed feet and a waterproof feather covering.

(a) Explain how each feature may aid survival in an aquatic environment. [2]

(b) Describe two requirements for either feature to be called an adaptive feature. [2]

Question 26

core · 4 marks · 0610_T18_P1_20261018_Q26

A beetle population contains inherited light and dark forms. Dark beetles are harder for predators to see on newly darkened ground.

(a) Explain why dark beetles may leave more offspring. [2]

(b) Explain how the frequency of dark beetles can change over generations. [2]

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Question 27

core · 4 marks · 0610_T18_P1_20261018_Q27

A breeder wants goats that produce more milk.

(a) Describe the first two stages of a selective-breeding programme. [2]

(b) Describe how the programme should continue. [2]

Question 28

supplement · 4 marks · 0610_T18_P1_20261018_Q28

Before treatment, a small fraction of bacteria resist an antibiotic.

(a) Explain the immediate effect of antibiotic treatment on the population. [2]

(b) Explain why the resistant fraction may increase after treatment. [2]

Question 29

supplement · 4 marks · 0610_T18_P1_20261018_Q29

A wild grass population survives repeated droughts. A farmer separately breeds grass selected for high seed yield.

(a) Explain how the wild population can become adapted to drought. [2]

(b) Contrast how reproductive success is determined in the two populations. [2]

Question 30

core · 4 marks · 0610_T18_P1_20261018_Q30

A fish species produces many more young than can survive in a pond.

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(a) Explain why this can lead to a struggle for survival. [2]

(b) Explain why this struggle does not make evolution a process of individual choice. [2]

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